

KITABU QUEST

Grade 4

SCIENCE AND TECHNOLOGY



- Clear lessons
- Worked examples
- Fun activities
- Real-life problems

KEN CBC aligned draft

GRADE

4

Title Page

KITABU QUEST Grade 4 Science and Technology

Kenya CBC content draft for review

Mascot: chameleon

This content-first draft was regenerated from Kitabu curriculum data using a topic-by-topic book plan. It is designed to help learners read, practise, discuss, create, and revise with confidence.

Cover artwork is intentionally deferred until all content has passed review and is marked ready-for-cover.

How To Use This Book

This book helps Grade 4 learners study Science and Technology through clear lessons, worked examples, activities, practice, and revision.

Use the inquiry questions to think first. Read the explanation. Try the guided example. Complete the activity. Answer the practice questions. Correct your work and ask for support where needed.

Observe, ask questions, test safely, record evidence, and explain findings.

Every unit has a local example, a misconception to avoid, success criteria, and a check for understanding.

Learning Skills In This Book

Each chapter builds knowledge, skills, values, creativity, communication, collaboration, critical thinking, digital literacy, and self-confidence.

Look for these page types: Learn, Example, Activity, Practice, Reflection, Home Link, and Review.

How to study well: read the goal first, underline new words, try the example before reading the answer, discuss with a partner, and correct your work after feedback. Do not skip the reflection questions. They help you notice what you understand and what needs more practice.

Table Of Contents

Use this table of contents to plan your study. Start with the first chapter, but return to earlier pages whenever you need revision.

1. 1.0 Living Things and their Environment
2. 1.1 Plants 1.0 Living Things and their Environment
3. 2.0 Matter 2.1 Properties of
4. 3.0 Force and Energy
5. 3.1 Force and its Effect 3.0 Force and

After each chapter, complete the chapter review before moving forward. At the end of the book, use the glossary, answer notes, and final project to revise the whole subject.

Chapter: Living Things and their Environment

In this chapter you will study Living Things and their Environment.

Inquiry starter: What do you already know about this topic from home, school, or your community?

Chapter units:

- 1.0 Living 1.3 Human Digestive System Things and their Environment
- 1.2 Animals

Before you begin, write two things you already know and one question you want to answer. As you study, collect examples from daily life. At the end of the chapter, return to your question and improve your answer using new vocabulary, examples, and evidence from the lessons.

Living 1.3 Human Digestive System Things and their Environment: Lesson Opener

Learning area: Living 1.3 Human Digestive System Things and their Environment

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Living 1.3 Human Digestive System Things and their Environment.

Curriculum link: This lesson follows the Grade 4 curriculum sequence and focuses on Living 1.3 Human Digestive System Things and their Environment.

By the end of this lesson sequence, you should be able to:

- identify parts of the human digestive system
- develop a plan of maintaining a healthy human digestive system
- appreciate the importance of a healthy human digestive system.

Inquiry questions:

- What makes up the digestive system?
- How is a healthy digestive system maintained?

Before reading, write one thing you already know and one question you want this lesson to answer.

Living 1.3 Human Digestive System Things and their Environment: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Living 1.3 Human Digestive System Things and their Environment.

Important idea: Living 1.3 Human Digestive System Things and their Environment.

Digestion changes food into small soluble substances the body can absorb. Mechanical digestion breaks food into smaller pieces. Chemical digestion uses enzymes to break large food molecules into smaller ones. A good explanation follows food through the mouth, oesophagus, stomach, small intestine, large intestine, and rectum.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 4, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

Evidence tool	How to use it in Living 1.3 Human Digestive System Things and their Environment
Labelled sketch	Name visible parts and show direction or change with arrows.
Observation table	Record facts before explaining them.
Safety note	Name the hazard and the safe action.
Conclusion	Begin with: The evidence shows...

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Living 1.3 Human Digestive System Things and their Environment
- Living
- Human
- Digestive
- System
- Things
- their
- Environment

Explain the idea in your own words before attempting the activities.

Living 1.3 Human Digestive System Things and their Environment: Worked Example

Investigation model for Living 1.3 Human Digestive System Things and their Environment: Trace what happens to food in the human digestive system.

Question: How do mechanical and chemical digestion work together?

Example: In the mouth, teeth break food into smaller pieces while saliva starts chemical digestion of starch. In the stomach, muscles churn food and gastric juice begins protein digestion. In the small intestine, enzymes complete digestion and digested food is absorbed.

Procedure:

1. Draw a labelled path: mouth, oesophagus, stomach, small intestine, large intestine, rectum.
2. Beside each part, write whether the main action is mechanical digestion, chemical digestion, absorption, or removal of waste.
3. Use arrows to show movement of food.
4. Add one enzyme or digestive juice where it is taught in class.
5. Explain why digestion is not complete in the mouth only.

Conclusion frame: Digestion needs both physical breakdown and chemical breakdown because _____.

Living 1.3 Human Digestive System Things and their Environment: Activity

Activity for Living 1.3 Human Digestive System Things and their Environment: Plan a safe observation or investigation.

Inquiry question: What makes up the digestive system?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Living 1.3 Human Digestive System Things and their Environment: Activity And Practice

Practice for Living 1.3 Human Digestive System Things and their Environment:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: identify parts of the human digestive system. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
2. Complete a science response for this outcome: develop a plan of maintaining a healthy human digestive system. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
3. Complete a science response for this outcome: appreciate the importance of a healthy human digestive system.. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Living 1.3 Human Digestive System Things and their Environment: Outcome Check

Outcome: identify parts of the human digestive system

Show mastery with topic evidence.

Task: Trace the food path through the digestive system, name mechanical and chemical digestion steps, and explain one adaptation of a digestive organ.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Living 1.3 Human Digestive System Things and their Environment, uses correct science vocabulary, and connects evidence to explanation.

Living 1.3 Human Digestive System Things and their Environment: Outcome Check

Outcome: develop a plan of maintaining a healthy human digestive system

Show mastery with topic evidence.

Task: Trace the food path through the digestive system, name mechanical and chemical digestion steps, and explain one adaptation of a digestive organ.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Living 1.3 Human Digestive System Things and their Environment, uses correct science vocabulary, and connects evidence to explanation.

Living 1.3 Human Digestive System Things and their Environment: Outcome Check

Outcome: appreciate the importance of a healthy human digestive system.

Show mastery with topic evidence.

Task: Trace the food path through the digestive system, name mechanical and chemical digestion steps, and explain one adaptation of a digestive organ.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Living 1.3 Human Digestive System Things and their Environment, uses correct science vocabulary, and connects evidence to explanation.

Animals: Lesson Opener

Learning area: Animals

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Animals.

Curriculum link: This lesson follows the Grade 4 curriculum sequence and focuses on Animals.

By the end of this lesson sequence, you should be able to:

- characteristics of animals as living things, search for information from print and non- vertebrates from invertebrates

Inquiry questions:

- Why are animals grouped as living things?

Before reading, write one thing you already know and one question you want this lesson to answer.

Animals: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Animals.

Important idea: Animals.

Science and technology learning starts with a question. A useful answer is built from evidence: observations, measurements, diagrams, models, tests, data tables, or design trials. Separate what you saw from what you think it means.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 4, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

| Evidence tool | How to use it in Animals |

| --- | --- |

| Labelled sketch | Name visible parts and show direction or change with arrows. |

| Observation table | Record facts before explaining them. |

| Safety note | Name the hazard and the safe action. |

| Conclusion | Begin with: The evidence shows... |

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Animals

Explain the idea in your own words before attempting the activities.

Animals: Worked Example

Investigation model for Animals: Move from question to evidence.

Question: What evidence would help explain this idea clearly?

Procedure:

1. State the exact question in one sentence.
2. Name the materials, diagram, data, model, or digital tool needed.
3. Decide what will be observed, measured, compared, or designed.
4. Record results in a table, labelled sketch, flow chart, or screenshot note.
5. Explain what the evidence means and name one limit of the investigation.

Evidence table:

Step	Evidence collected	What it shows
---	---	---
Observe or test	specific detail	explanation

Conclusion frame: The evidence supports this answer because _____.

Animals: Activity

Activity for Animals: Plan a safe observation or investigation.

Inquiry question: Why are animals grouped as living things?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Animals: Activity And Practice

Practice for Animals:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: characteristics of animals as living things, search for information from print and non- vertebrates from invertebrates. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Animals: Outcome Check

Outcome: characteristics of animals as living things, search for information from print and non-vertebrates from invertebrates

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Animals, uses correct science vocabulary, and connects evidence to explanation.

Living Things and their Environment: Chapter Review

Review the chapter carefully.

1. Write five key words from this chapter and explain each one.
2. Choose two units and compare what you learned.
3. Create one poster, chart, dialogue, model, map, or worked solution that teaches this chapter.
4. Answer one inquiry question again. Improve your first answer using what you now know.
5. Rate your confidence: high, medium, or needs practice.

Chapter: Plants 1.0 Living Things and their Environment

In this chapter you will study Plants 1.0 Living Things and their Environment.

Inquiry starter: What do you already know about this topic from home, school, or your community?

Chapter units:

- 1.1 Plants 1.0 Living Things and their Environment

Before you begin, write two things you already know and one question you want to answer. As you study, collect examples from daily life. At the end of the chapter, return to your question and improve your answer using new vocabulary, examples, and evidence from the lessons.

Plants 1.0 Living Things and their Environment: Lesson Opener

Learning area: Plants 1.0 Living Things and their Environment

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Plants 1.0 Living Things and their Environment.

Curriculum link: This lesson follows the Grade 4 curriculum sequence and focuses on Plants 1.0 Living Things and their Environment.

By the end of this lesson sequence, you should be able to:

- be print materials on characteristics of plants as able to: living things and discuss with peers

Inquiry questions:

- Why are plants grouped as living things?

Before reading, write one thing you already know and one question you want this lesson to answer.

Plants 1.0 Living Things and their Environment: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Plants 1.0 Living Things and their Environment.

Important idea: Plants 1.0 Living Things and their Environment.

Environmental science connects observation with care for people and living things. Record place, date, conditions, and evidence. Then explain the pattern and propose a practical improvement that protects health, safety, food, water, or the environment.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 4, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

| Evidence tool | How to use it in Plants 1.0 Living Things and their Environment |

| --- | --- |

| Labelled sketch | Name visible parts and show direction or change with arrows. |

| Observation table | Record facts before explaining them. |

| Safety note | Name the hazard and the safe action. |

| Conclusion | Begin with: The evidence shows... |

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Plants 1.0 Living Things and their Environment
- Plants
- Living
- Things
- their
- Environment

Explain the idea in your own words before attempting the activities.

Plants 1.0 Living Things and their Environment: Worked Example

Investigation model for Plants 1.0 Living Things and their Environment: Use field observation carefully.

Question: What evidence in the local environment shows the pattern or problem?

Example: A school garden with dry soil, wilting leaves, and cracks near the roots shows that plants need steady water and soil cover to reduce moisture loss.

Procedure:

1. Choose a safe observation point in the school compound, garden, home area, or community.
2. Record date, weather, place, and what you can see without disturbing living things.
3. Draw a labelled sketch or complete a three-column observation table.
4. Separate facts from explanations.
5. Suggest one practical improvement that protects health, safety, food, water, or the environment.

Conclusion frame: My observations show _____ because the evidence was _____.

Plants 1.0 Living Things and their Environment: Activity

Activity for Plants 1.0 Living Things and their Environment: Plan a safe observation or investigation.

Inquiry question: Why are plants grouped as living things?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Plants 1.0 Living Things and their Environment: Activity And Practice

Practice for Plants 1.0 Living Things and their Environment:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: be print materials on characteristics of plants as able to: living things and discuss with peers. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Plants 1.0 Living Things and their Environment: Outcome Check

Outcome: be print materials on characteristics of plants as able to: living things and discuss with peers

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Plants 1.0 Living Things and their Environment, uses correct science vocabulary, and connects evidence to explanation.

Plants 1.0 Living Things and their Environment: Chapter Review

Review the chapter carefully.

1. Write five key words from this chapter and explain each one.
2. Choose two units and compare what you learned.
3. Create one poster, chart, dialogue, model, map, or worked solution that teaches this chapter.
4. Answer one inquiry question again. Improve your first answer using what you now know.
5. Rate your confidence: high, medium, or needs practice.

Chapter: Matter 2.1 Properties of

In this chapter you will study Matter 2.1 Properties.

Inquiry starter: What do you already know about this topic from home, school, or your community?

Chapter units:

- 2.0 Matter 2.1 Properties

Before you begin, write two things you already know and one question you want to answer. As you study, collect examples from daily life. At the end of the chapter, return to your question and improve your answer using new vocabulary, examples, and evidence from the lessons.

Matter 2.1 Properties: categorise substances in the environment into the three: Lesson Opener

Learning area: Matter 2.1 Properties: categorise substances in the environment into the three

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Matter 2.1 Properties: categorise substances in the environment into the three.

Curriculum link: This lesson follows the Grade 4 curriculum sequence and focuses on Matter 2.1 Properties: categorise substances in the environment into the three.

By the end of this lesson sequence, you should be able to:

- categorise substances in the environment into the three states of matter
- describe the properties of the three states of matter
- appreciate the importance of the different states of matter in day-to-day life.

Inquiry questions:

- What are the properties of matter?
- How is solid waste dangerous to the environment?
- Why is it important to conserve water?

Before reading, write one thing you already know and one question you want this lesson to answer.

Matter 2.1 Properties: categorise substances in the environment into the three: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Matter 2.1 Properties: categorise substances in the environment into the three.

Important idea: Matter 2.1 Properties: categorise substances in the environment into the three.

Materials can be described by observable properties such as state, texture, solubility, magnetism, colour, and reaction. Choose a separation or testing method because it matches a property, not because the method sounds familiar.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 4, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

| Evidence tool | How to use it in Matter 2.1 Properties: categorise substances in the environment into the three |

| --- | --- |

| Labelled sketch | Name visible parts and show direction or change with arrows. |

| Observation table | Record facts before explaining them. |

| Safety note | Name the hazard and the safe action. |

| Conclusion | Begin with: The evidence shows... |

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Matter 2.1 Properties: categorise substances in the environment into the three
- Matter 2.1 Properties
- Matter
- Properties
- categorise
- substances
- environment
- three

Explain the idea in your own words before attempting the activities.

Matter 2.1 Properties: categorise substances in the environment into the three: Worked Example

Investigation model for Matter 2.1 Properties: categorise substances in the environment into the three: Compare material properties.

Question: Which property helps us identify or separate the materials?

Example: A mixture of sand and salt can be separated because salt dissolves in water but sand does not. The sand can be filtered out, and salt can be recovered from the solution by evaporation.

Procedure:

1. List the materials and their visible properties.
2. Predict which property will be useful: size, solubility, magnetism, boiling point, texture, or colour change.
3. Choose a safe method that matches the property.
4. Record what happens before and after each step.
5. Explain why the method worked instead of only naming the method.

Conclusion frame: The useful property was _____, so the best method was _____.

Matter 2.1 Properties: categorise substances in the environment into the three: Activity

Activity for Matter 2.1 Properties: categorise substances in the environment into the three: Plan a safe observation or investigation.

Inquiry question: What are the properties of matter?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Matter 2.1 Properties: categorise substances in the environment into the three: Activity And Practice

Practice for Matter 2.1 Properties: categorise substances in the environment into the three:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: categorise substances in the environment into the three states of matter. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
2. Complete a science response for this outcome: describe the properties of the three states of matter. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
3. Complete a science response for this outcome: appreciate the importance of the different states of matter in day-to-day life.. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Matter 2.1 Properties: categorise substances in the environment into the three: Outcome Check

Outcome: categorise substances in the environment into the three states of matter

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Matter 2.1 Properties: categorise substances in the environment into the three, uses correct science vocabulary, and connects evidence to explanation.

Matter 2.1 Properties: categorise substances in the environment into the three: Outcome Check

Outcome: describe the properties of the three states of matter

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Matter 2.1 Properties: categorise substances in the environment into the three, uses correct science vocabulary, and connects evidence to explanation.

Matter 2.1 Properties: categorise substances in the environment into the three: Outcome Check

Outcome: appreciate the importance of the different states of matter in day-to-day life.

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Matter 2.1 Properties: categorise substances in the environment into the three, uses correct science vocabulary, and connects evidence to explanation.

Matter 2.1 Properties: classify solid waste into decomposable and non- decomposable: Lesson Opener

Learning area: Matter 2.1 Properties: classify solid waste into decomposable and non- decomposable

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Matter 2.1 Properties: classify solid waste into decomposable and non- decomposable.

Curriculum link: This lesson follows the Grade 4 curriculum sequence and focuses on Matter 2.1 Properties: classify solid waste into decomposable and non- decomposable.

By the end of this lesson sequence, you should be able to:

- classify solid waste into decomposable and non- decomposable
- describe the dangers of solid waste to the environment
- apply appropriate methods to manage solid waste in the environment, Hint: Include common waste in school

Inquiry questions:

- How is solid waste dangerous to the environment?
- Why is it important to conserve water?

Before reading, write one thing you already know and one question you want this lesson to answer.

Matter 2.1 Properties: classify solid waste into decomposable and non- decomposable: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Matter 2.1 Properties: classify solid waste into decomposable and non- decomposable.

Important idea: Matter 2.1 Properties: classify solid waste into decomposable and non- decomposable.

Science and technology learning starts with a question. A useful answer is built from evidence: observations, measurements, diagrams, models, tests, data tables, or design trials. Separate what you saw from what you think it means.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 4, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

| Evidence tool | How to use it in Matter 2.1 Properties: classify solid waste into decomposable and non- decomposable |

| --- | --- |

| Labelled sketch | Name visible parts and show direction or change with arrows. |

| Observation table | Record facts before explaining them. |

| Safety note | Name the hazard and the safe action. |

| Conclusion | Begin with: The evidence shows... |

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Matter 2.1 Properties: classify solid waste into decomposable and non- decomposable
- Matter 2.1 Properties
- Matter
- Properties
- classify
- solid
- waste
- decomposable

Explain the idea in your own words before attempting the activities.

Matter 2.1 Properties: classify solid waste into decomposable and non- decomposable: Worked Example

Investigation model for Matter 2.1 Properties: classify solid waste into decomposable and non- decomposable: Compare states of matter.

Question: How do solids, liquids, and gases behave differently?

Example comparison:

State	Shape	Volume	Example
---	---	---	---
Solid	fixed	fixed	stone
Liquid	takes container shape	fixed	water
Gas	fills space	not fixed	air

Procedure:

1. Observe one safe solid, liquid, and gas example.
2. Compare shape and volume.
3. Describe particle arrangement using simple spacing language.
4. Explain one change of state such as melting, freezing, evaporation, or condensation.
5. Record what changed and what stayed the same.

Conclusion frame: Matter changes state when _____.

Matter 2.1 Properties: classify solid waste into decomposable and non- decomposable: Activity

Activity for Matter 2.1 Properties: classify solid waste into decomposable and non- decomposable: Plan a safe observation or investigation.

Inquiry question: How is solid waste dangerous to the environment?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Matter 2.1 Properties: classify solid waste into decomposable and non- decomposable: Activity And Practice

Practice for Matter 2.1 Properties: classify solid waste into decomposable and non- decomposable:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: classify solid waste into decomposable and non-decomposable. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
2. Complete a science response for this outcome: describe the dangers of solid waste to the environment. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
3. Complete a science response for this outcome: apply appropriate methods to manage solid waste in the environment, Hint: Include common waste in school. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Matter 2.1 Properties: classify solid waste into decomposable and non- decomposable: Outcome Check

Outcome: classify solid waste into decomposable and non- decomposable

Show mastery with topic evidence.

Task: Complete a table comparing solid, liquid, and gas by shape, volume, particle arrangement, and one example, then explain one change of state. Answer guide: solids keep shape, liquids keep volume but take container shape, gases spread to fill space; melting changes solid to liquid.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Matter 2.1 Properties: classify solid waste into decomposable and non- decomposable, uses correct science vocabulary, and connects evidence to explanation.

Matter 2.1 Properties: classify solid waste into decomposable and non- decomposable: Outcome Check

Outcome: describe the dangers of solid waste to the environment

Show mastery with topic evidence.

Task: Complete a table comparing solid, liquid, and gas by shape, volume, particle arrangement, and one example, then explain one change of state. Answer guide: solids keep shape, liquids keep volume but take container shape, gases spread to fill space; melting changes solid to liquid.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Matter 2.1 Properties: classify solid waste into decomposable and non- decomposable, uses correct science vocabulary, and connects evidence to explanation.

Matter 2.1 Properties: classify solid waste into decomposable and non- decomposable: Outcome Check

Outcome: apply appropriate methods to manage solid waste in the environment, Hint: Include common waste in school

Show mastery with topic evidence.

Task: Complete a table comparing solid, liquid, and gas by shape, volume, particle arrangement, and one example, then explain one change of state. Answer guide: solids keep shape, liquids keep volume but take container shape, gases spread to fill space; melting changes solid to liquid.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Matter 2.1 Properties: classify solid waste into decomposable and non- decomposable, uses correct science vocabulary, and connects evidence to explanation.

Matter 2.1 Properties: environment such as kitchen, animal waste, need: Lesson Opener

Learning area: Matter 2.1 Properties: environment such as kitchen, animal waste, need

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Matter 2.1 Properties: environment such as kitchen, animal waste, need.

Curriculum link: This lesson follows the Grade 4 curriculum sequence and focuses on Matter 2.1 Properties: environment such as kitchen, animal waste, need.

By the end of this lesson sequence, you should be able to:

- appreciate the environment such as kitchen, animal waste, need for proper plastics, e-waste, metals and glasses
- Project: Prepare posters to sensitise the community (school and home) on the importance of water conservation.

Inquiry questions:

- Why is it important to conserve water?

Before reading, write one thing you already know and one question you want this lesson to answer.

Matter 2.1 Properties: environment such as kitchen, animal waste, need: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Matter 2.1 Properties: environment such as kitchen, animal waste, need.

Important idea: Matter 2.1 Properties: environment such as kitchen, animal waste, need.

Science and technology learning starts with a question. A useful answer is built from evidence: observations, measurements, diagrams, models, tests, data tables, or design trials. Separate what you saw from what you think it means.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 4, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

| Evidence tool | How to use it in Matter 2.1 Properties: environment such as kitchen, animal waste, need |

| --- | --- |

| Labelled sketch | Name visible parts and show direction or change with arrows. |

| Observation table | Record facts before explaining them. |

| Safety note | Name the hazard and the safe action. |

| Conclusion | Begin with: The evidence shows... |

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Matter 2.1 Properties: environment such as kitchen, animal waste, need
- Matter 2.1 Properties
- Matter
- Properties
- environment
- kitchen
- animal
- waste

Explain the idea in your own words before attempting the activities.

Matter 2.1 Properties: environment such as kitchen, animal waste, need: Worked Example

Observation model for Matter 2.1 Properties: environment such as kitchen, animal waste, need: Classify animals by visible features.

Question: What features help us group animals correctly?

Example: A fish has scales, fins, and gills, so it is adapted for living in water. A bird has feathers, wings, and a beak, so those features help with movement, warmth, feeding, or protection.

Procedure:

1. Choose three familiar animals.
2. Record body covering, movement, feeding structure, habitat, and one adaptation.
3. Group the animals using visible evidence.
4. Explain why one feature supports survival.
5. Avoid grouping by guesses or likes.

Conclusion frame: The evidence shows this animal belongs to _____ because _____.

Matter 2.1 Properties: environment such as kitchen, animal waste, need: Activity

Activity for Matter 2.1 Properties: environment such as kitchen, animal waste, need: Plan a safe observation or investigation.

Inquiry question: Why is it important to conserve water?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Matter 2.1 Properties: environment such as kitchen, animal waste, need: Activity And Practice

Practice for Matter 2.1 Properties: environment such as kitchen, animal waste, need:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: appreciate the environment such as kitchen, animal waste, need for proper plastics, e-waste, metals and glasses. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
2. Complete a science response for this outcome: Project: Prepare posters to sensitise the community (school and home) on the importance of water conservation.. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Matter 2.1 Properties: environment such as kitchen, animal waste, need: Outcome Check

Outcome: appreciate the environment such as kitchen, animal waste, need for proper plastics, e-waste, metals and glasses

Show mastery with topic evidence.

Task: Classify three animals by observable features, habitat, movement, or body covering, then explain one adaptation. Answer guide: name the animal, give visible evidence such as feathers/scales/legs/body parts, and connect the feature to survival.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Matter 2.1 Properties: environment such as kitchen, animal waste, need, uses correct science vocabulary, and connects evidence to explanation.

Matter 2.1 Properties: environment such as kitchen, animal waste, need: Outcome Check

Outcome: Project: Prepare posters to sensitise the community (school and home) on the importance of water conservation.

Show mastery with topic evidence.

Task: Classify three animals by observable features, habitat, movement, or body covering, then explain one adaptation. Answer guide: name the animal, give visible evidence such as feathers/scales/legs/body parts, and connect the feature to survival.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Matter 2.1 Properties: environment such as kitchen, animal waste, need, uses correct science vocabulary, and connects evidence to explanation.

Matter 2.1 Properties of: Chapter Review

Review the chapter carefully.

1. Write five key words from this chapter and explain each one.
2. Choose two units and compare what you learned.
3. Create one poster, chart, dialogue, model, map, or worked solution that teaches this chapter.
4. Answer one inquiry question again. Improve your first answer using what you now know.
5. Rate your confidence: high, medium, or needs practice.

Chapter: Force and Energy

In this chapter you will study Force and Energy.

Inquiry starter: What do you already know about this topic from home, school, or your community?

Chapter units:

- 3.0 Force 3.3 Heat
- 3.2 Light

Before you begin, write two things you already know and one question you want to answer. As you study, collect examples from daily life. At the end of the chapter, return to your question and improve your answer using new vocabulary, examples, and evidence from the lessons.

Force 3.3 Heat: Lesson Opener

Learning area: Force 3.3 Heat

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Force 3.3 Heat.

Curriculum link: This lesson follows the Grade 4 curriculum sequence and focuses on Force 3.3 Heat.

By the end of this lesson sequence, you should be able to:

- Project: Use locally available materials to make kitchen gloves.

Inquiry questions:

- What are the safety measures observed when using heat?

Before reading, write one thing you already know and one question you want this lesson to answer.

Force 3.3 Heat: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Force 3.3 Heat.

Important idea: Force 3.3 Heat.

Physical science looks for patterns in energy, forces, and materials. Keep one condition changing at a time, observe the effect, and use measured evidence where possible. A correct explanation names the cause, the change observed, and the reason the change happened.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 4, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

Evidence tool	How to use it in Force 3.3 Heat
Labelled sketch	Name visible parts and show direction or change with arrows.
Observation table	Record facts before explaining them.
Safety note	Name the hazard and the safe action.
Conclusion	Begin with: The evidence shows...

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Force 3.3 Heat
- Force

Explain the idea in your own words before attempting the activities.

Force 3.3 Heat: Worked Example

Investigation model for Force 3.3 Heat: Test one energy or circuit condition safely.

Question: What changes when one condition is changed?

Example: In a simple circuit, a bulb lights only when there is a complete path from one battery terminal, through the bulb, and back to the other terminal.

Procedure:

1. Name the variable you will change, such as number of cells, wire length, material, distance, or surface.
2. Name what you will observe or measure.
3. Keep the other conditions the same.
4. Record results in a table with at least three trials or comparisons.
5. Write a conclusion that mentions the pattern in the results.

Safety rule: Use only approved school materials and ask before connecting any electrical device.

Force 3.3 Heat: Activity

Activity for Force 3.3 Heat: Plan a safe observation or investigation.

Inquiry question: What are the safety measures observed when using heat?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Force 3.3 Heat: Activity And Practice

Practice for Force 3.3 Heat:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: Project: Use locally available materials to make kitchen gloves.. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Force 3.3 Heat: Outcome Check

Outcome: Project: Use locally available materials to make kitchen gloves.

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Force 3.3 Heat, uses correct science vocabulary, and connects evidence to explanation.

Light: sources of light in nature: Lesson Opener

Learning area: Light: sources of light in nature

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Light: sources of light in nature.

Curriculum link: This lesson follows the Grade 4 curriculum sequence and focuses on Light: sources of light in nature.

By the end of this lesson sequence, you should be able to:

- identify the sources of light in nature
- describe ways of lighting a house in day-to-day life
- explain the uses of light in day-to-day life

Inquiry questions:

- Why is light important in day-to-day life?

Before reading, write one thing you already know and one question you want this lesson to answer.

Light: sources of light in nature: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Light: sources of light in nature.

Important idea: Light: sources of light in nature.

Physical science looks for patterns in energy, forces, and materials. Keep one condition changing at a time, observe the effect, and use measured evidence where possible. A correct explanation names the cause, the change observed, and the reason the change happened.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 4, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

Evidence tool	How to use it in Light: sources of light in nature
Labelled sketch	Name visible parts and show direction or change with arrows.
Observation table	Record facts before explaining them.
Safety note	Name the hazard and the safe action.
Conclusion	Begin with: The evidence shows...

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Light: sources of light in nature
- Light
- sources
- nature

Explain the idea in your own words before attempting the activities.

Light: sources of light in nature: Worked Example

Investigation model for Light: sources of light in nature: Test one energy or circuit condition safely.

Question: What changes when one condition is changed?

Example: In a simple circuit, a bulb lights only when there is a complete path from one battery terminal, through the bulb, and back to the other terminal.

Procedure:

1. Name the variable you will change, such as number of cells, wire length, material, distance, or surface.
2. Name what you will observe or measure.
3. Keep the other conditions the same.
4. Record results in a table with at least three trials or comparisons.
5. Write a conclusion that mentions the pattern in the results.

Safety rule: Use only approved school materials and ask before connecting any electrical device.

Light: sources of light in nature: Activity

Activity for Light: sources of light in nature: Plan a safe observation or investigation.

Inquiry question: Why is light important in day-to-day life?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Light: sources of light in nature: Activity And Practice

Practice for Light: sources of light in nature:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: identify the sources of light in nature. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
2. Complete a science response for this outcome: describe ways of lighting a house in day- to- day life. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
3. Complete a science response for this outcome: explain the uses of light in day-to-day life. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Light: sources of light in nature: Outcome Check

Outcome: identify the sources of light in nature

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Light: sources of light in nature, uses correct science vocabulary, and connects evidence to explanation.

Light: sources of light in nature: Outcome Check

Outcome: describe ways of lighting a house in day- to- day life

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Light: sources of light in nature, uses correct science vocabulary, and connects evidence to explanation.

Light: sources of light in nature: Outcome Check

Outcome: explain the uses of light in day-to-day life

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Light: sources of light in nature, uses correct science vocabulary, and connects evidence to explanation.

Light: applications of light in day-to-day life.: Lesson Opener

Learning area: Light: applications of light in day-to-day life.

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Light: applications of light in day-to-day life..

Curriculum link: This lesson follows the Grade 4 curriculum sequence and focuses on Light: applications of light in day-to-day life..

By the end of this lesson sequence, you should be able to:

- appreciate the applications of light in day-to-day life.

Inquiry questions:

- Why is light important in day-to-day life?

Before reading, write one thing you already know and one question you want this lesson to answer.

Light: applications of light in day-to-day life.: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Light: applications of light in day-to-day life..

Important idea: Light: applications of light in day-to-day life..

Physical science looks for patterns in energy, forces, and materials. Keep one condition changing at a time, observe the effect, and use measured evidence where possible. A correct explanation names the cause, the change observed, and the reason the change happened.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 4, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

Evidence tool	How to use it in Light: applications of light in day-to-day life.
Labelled sketch	Name visible parts and show direction or change with arrows.
Observation table	Record facts before explaining them.
Safety note	Name the hazard and the safe action.
Conclusion	Begin with: The evidence shows...

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Light: applications of light in day-to-day life.
- Light
- applications
- day-to-day
- life.

Explain the idea in your own words before attempting the activities.

Light: applications of light in day-to-day life.: Worked Example

Investigation model for Light: applications of light in day-to-day life.: Test one energy or circuit condition safely.

Question: What changes when one condition is changed?

Example: In a simple circuit, a bulb lights only when there is a complete path from one battery terminal, through the bulb, and back to the other terminal.

Procedure:

1. Name the variable you will change, such as number of cells, wire length, material, distance, or surface.
2. Name what you will observe or measure.
3. Keep the other conditions the same.
4. Record results in a table with at least three trials or comparisons.
5. Write a conclusion that mentions the pattern in the results.

Safety rule: Use only approved school materials and ask before connecting any electrical device.

Light: applications of light in day-to-day life.: Activity

Activity for Light: applications of light in day-to-day life.: Plan a safe observation or investigation.

Inquiry question: Why is light important in day-to-day life?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Light: applications of light in day-to-day life.: Activity And Practice

Practice for Light: applications of light in day-to-day life.:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: appreciate the applications of light in day-to-day life.. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Light: applications of light in day-to-day life.: Outcome Check

Outcome: appreciate the applications of light in day-to-day life.

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Light: applications of light in day-to-day life., uses correct science vocabulary, and connects evidence to explanation.

Force and Energy: Chapter Review

Review the chapter carefully.

1. Write five key words from this chapter and explain each one.
2. Choose two units and compare what you learned.
3. Create one poster, chart, dialogue, model, map, or worked solution that teaches this chapter.
4. Answer one inquiry question again. Improve your first answer using what you now know.
5. Rate your confidence: high, medium, or needs practice.

Chapter: Force and its Effect 3.0 Force and

In this chapter you will study Force and its Effect 3.0 Force.

Inquiry starter: What do you already know about this topic from home, school, or your community?

Chapter units:

- 3.1 Force and its Effect 3.0 Force

Before you begin, write two things you already know and one question you want to answer. As you study, collect examples from daily life. At the end of the chapter, return to your question and improve your answer using new vocabulary, examples, and evidence from the lessons.

Force and its Effect 3.0 Force: types of forces in nature: Lesson Opener

Learning area: Force and its Effect 3.0 Force: types of forces in nature

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Force and its Effect 3.0 Force: types of forces in nature.

Curriculum link: This lesson follows the Grade 4 curriculum sequence and focuses on Force and its Effect 3.0 Force: types of forces in nature.

By the end of this lesson sequence, you should be able to:

- describe types of forces in nature
- demonstrate the carry out activities to demonstrate the effects of force on objects in nature, use
- describe the uses of force in daily life, discuss with peers ways of minimising

Inquiry questions:

- How does force affect objects?

Before reading, write one thing you already know and one question you want this lesson to answer.

Force and its Effect 3.0 Force: types of forces in nature: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Force and its Effect 3.0 Force: types of forces in nature.

Important idea: Force and its Effect 3.0 Force: types of forces in nature.

Forces and motion are studied by measuring how objects start, stop, speed up, slow down, or change direction. A fair test changes one condition at a time and records distance, time, speed, force, or pressure accurately.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 4, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

Evidence tool	How to use it in Force and its Effect 3.0 Force: types of forces in nature
Labelled sketch	Name visible parts and show direction or change with arrows.
Observation table	Record facts before explaining them.
Safety note	Name the hazard and the safe action.
Conclusion	Begin with: The evidence shows...

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Force and its Effect 3.0 Force: types of forces in nature
- Force and its Effect 3.0 Force
- Force
- Effect
- types
- forces
- nature

Explain the idea in your own words before attempting the activities.

Force and its Effect 3.0 Force: types of forces in nature: Worked Example

Investigation model for Force and its Effect 3.0 Force: types of forces in nature: Measure a change in motion.

Question: How does the force or surface affect the movement?

Example: A toy car travels farther on a smooth tile than on rough soil because the rough surface produces more friction.

Procedure:

1. Choose one object to move.
2. Change only one condition, such as surface, load, slope, or pushing force.
3. Measure distance or time in the same way each trial.
4. Repeat the test and record the results.
5. Use the words force, motion, friction, speed, or pressure correctly in the conclusion.

Conclusion frame: When _____ changed, _____ happened because _____.

Force and its Effect 3.0 Force: types of forces in nature: Activity

Activity for Force and its Effect 3.0 Force: types of forces in nature: Plan a safe observation or investigation.

Inquiry question: How does force affect objects?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Force and its Effect 3.0 Force: types of forces in nature: Activity And Practice

Practice for Force and its Effect 3.0 Force: types of forces in nature:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: describe types of forces in nature. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
2. Complete a science response for this outcome: demonstrate the carry out activities to demonstrate the effects of force on objects in nature, use. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.
3. Complete a science response for this outcome: describe the uses of force in daily life, discuss with peers ways of minimising. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Force and its Effect 3.0 Force: types of forces in nature: Outcome Check

Outcome: describe types of forces in nature

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Force and its Effect 3.0 Force: types of forces in nature, uses correct science vocabulary, and connects evidence to explanation.

Force and its Effect 3.0 Force: types of forces in nature: Outcome Check

Outcome: demonstrate the carry out activities to demonstrate the effects of force on objects in nature, use

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
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5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Force and its Effect 3.0 Force: types of forces in nature, uses correct science vocabulary, and connects evidence to explanation.

Force and its Effect 3.0 Force: types of forces in nature: Outcome Check

Outcome: describe the uses of force in daily life, discuss with peers ways of minimising

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
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4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Force and its Effect 3.0 Force: types of forces in nature, uses correct science vocabulary, and connects evidence to explanation.

Force and its Effect 3.0 Force: importance of force in day-to-day life. discuss: Lesson Opener

Learning area: Force and its Effect 3.0 Force: importance of force in day-to-day life. discuss

Start here: a safe observation at home, school, garden, workshop, health club, or local environment linked to Force and its Effect 3.0 Force: importance of force in day-to-day life. discuss.

Curriculum link: This lesson follows the Grade 4 curriculum sequence and focuses on Force and its Effect 3.0 Force: importance of force in day-to-day life. discuss.

By the end of this lesson sequence, you should be able to:

- appreciate the importance of force in day-to-day life. discuss the applications of force of friction

Inquiry questions:

- How does force affect objects?

Before reading, write one thing you already know and one question you want this lesson to answer.

Force and its Effect 3.0 Force: importance of force in day-to-day life. discuss: Learn

Start with this real situation: a safe observation at home, school, garden, workshop, health club, or local environment linked to Force and its Effect 3.0 Force: importance of force in day-to-day life. discuss.

Important idea: Force and its Effect 3.0 Force: importance of force in day-to-day life. discuss.

Forces and motion are studied by measuring how objects start, stop, speed up, slow down, or change direction. A fair test changes one condition at a time and records distance, time, speed, force, or pressure accurately.

A common mistake is writing guesses as facts without observing, testing safely, or recording evidence. Avoid it by separating what you saw from what you think it means.

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

For Grade 4, use labelled sketches, tables, and short evidence-based explanations.

Inline visual support:

| Evidence tool | How to use it in Force and its Effect 3.0 Force: importance of force in day-to-day life. discuss |

| --- | --- |

| Labelled sketch | Name visible parts and show direction or change with arrows. |

| Observation table | Record facts before explaining them. |

| Safety note | Name the hazard and the safe action. |

| Conclusion | Begin with: The evidence shows... |

Method for Science and Technology: Observe, ask questions, test safely, record evidence, and explain findings.

Key words:

- Force and its Effect 3.0 Force: importance of force in day-to-day life. discuss
- Force and its Effect 3.0 Force
- Force
- Effect
- day-to-day
- life.
- discuss

Explain the idea in your own words before attempting the activities.

Force and its Effect 3.0 Force: importance of force in day-to-day life. discuss: Worked Example

Investigation model for Force and its Effect 3.0 Force: importance of force in day-to-day life. discuss: Measure a change in motion.

Question: How does the force or surface affect the movement?

Example: A toy car travels farther on a smooth tile than on rough soil because the rough surface produces more friction.

Procedure:

1. Choose one object to move.
2. Change only one condition, such as surface, load, slope, or pushing force.
3. Measure distance or time in the same way each trial.
4. Repeat the test and record the results.
5. Use the words force, motion, friction, speed, or pressure correctly in the conclusion.

Conclusion frame: When _____ changed, _____ happened because _____.

Force and its Effect 3.0 Force: importance of force in day-to-day life. discuss: Activity

Activity for Force and its Effect 3.0 Force: importance of force in day-to-day life. discuss: Plan a safe observation or investigation.

Inquiry question: How does force affect objects?

1. State the question you want to answer.
2. List safe materials and one safety rule.
3. Draw a labelled setup or observation table before starting.
4. Record at least three observations. Do not write guesses as facts.
5. Compare evidence with a partner and discuss what the evidence means.
6. Write a short conclusion that begins, "The evidence shows..."

Success criteria:

- observation is specific
- safety is considered
- conclusion uses evidence

Home link: Look for a safe related example at home or in the community. Record what you observed without disturbing people, animals, plants, or devices.

Force and its Effect 3.0 Force: importance of force in day-to-day life. discuss: Activity And Practice

Practice for Force and its Effect 3.0 Force: importance of force in day-to-day life. discuss:

Use this routine for every answer: question, materials or evidence, method, observation, conclusion, and safety.

1. Complete a science response for this outcome: appreciate the importance of force in day-to-day life. discuss the applications of force of friction. Include the key concept, labelled diagram/table/model where useful, evidence or expected observation, safety note, and conclusion.

Correction check:

- observation is specific
- safety is considered
- conclusion uses evidence

Reflection: Which part of your answer is evidence? Which part is explanation?

Home link: Observe a safe related example at home or in the community. Record facts only, without disturbing people, animals, plants, or devices.

Force and its Effect 3.0 Force: importance of force in day-to-day life. discuss: Outcome Check

Outcome: appreciate the importance of force in day-to-day life. discuss the applications of force of friction

Show mastery with topic evidence.

Task: Use the topic-specific evidence tool: define the key concept, complete a labelled table or diagram with at least three accurate details, add one expected observation or example, and write a conclusion that directly answers the outcome.

Steps:

1. Define the key term or process in one accurate sentence.
2. Complete the diagram, table, model, comparison, calculation, or investigation plan required by the task.
3. Add labels, units, variables, or safety notes where they matter.
4. Write a conclusion that uses the words "The evidence shows..." or "The model shows..."
5. Name one common misconception and correct it.

Evidence of mastery: your work answers the exact outcome in Force and its Effect 3.0 Force: importance of force in day-to-day life. discuss, uses correct science vocabulary, and connects evidence to explanation.

Force and its Effect 3.0 Force and: Chapter Review

Review the chapter carefully.

1. Write five key words from this chapter and explain each one.
2. Choose two units and compare what you learned.
3. Create one poster, chart, dialogue, model, map, or worked solution that teaches this chapter.
4. Answer one inquiry question again. Improve your first answer using what you now know.
5. Rate your confidence: high, medium, or needs practice.

Living 1.3 Human Digestive System Things and their Environment: Review Clinic 1

Use this review clinic to strengthen a real unit from the book.

Practice context: home or school example connected to Living 1.3 Human Digestive System Things and their Environment.

1. Re-read the lesson on Living 1.3 Human Digestive System Things and their Environment and underline the main idea.
2. Write a five-sentence or five-step summary.
3. Make one question that tests knowledge.
4. Make one question that tests skill, reasoning, or application.
5. Solve or answer both questions.
6. Check your work against this criterion: observation is specific.
7. Exchange your questions with a partner and discuss corrections.

Challenge: Describe a real-life task from your home, school, farm, market, field, library, or community that uses this idea.

Animals: Review Clinic 2

Use this review clinic to strengthen a real unit from the book.

Practice context: county or community example connected to Animals.

1. Re-read the lesson on Animals and underline the main idea.
2. Write a five-sentence or five-step summary.
3. Make one question that tests knowledge.
4. Make one question that tests skill, reasoning, or application.
5. Solve or answer both questions.
6. Check your work against this criterion: safety is considered.
7. Exchange your questions with a partner and discuss corrections.

Challenge: Describe a real-life task from your home, school, farm, market, field, library, or community that uses this idea.

Plants 1.0 Living Things and their Environment: Review Clinic 3

Use this review clinic to strengthen a real unit from the book.

Practice context: partner discussion about Plants 1.0 Living Things and their Environment.

1. Re-read the lesson on Plants 1.0 Living Things and their Environment and underline the main idea.
2. Write a five-sentence or five-step summary.
3. Make one question that tests knowledge.
4. Make one question that tests skill, reasoning, or application.
5. Solve or answer both questions.
6. Check your work against this criterion: conclusion uses evidence.
7. Exchange your questions with a partner and discuss corrections.

Challenge: Describe a real-life task from your home, school, farm, market, field, library, or community that uses this idea.

Matter 2.1 Properties: categorise substances in the environment into the three: Review Clinic 4

Use this review clinic to strengthen a real unit from the book.

Practice context: small project or observation linked to Matter 2.1 Properties: categorise substances in the environment into the three.

1. Re-read the lesson on Matter 2.1 Properties: categorise substances in the environment into the three and underline the main idea.
2. Write a five-sentence or five-step summary.
3. Make one question that tests knowledge.
4. Make one question that tests skill, reasoning, or application.
5. Solve or answer both questions.
6. Check your work against this criterion: observation is specific.
7. Exchange your questions with a partner and discuss corrections.

Challenge: Describe a real-life task from your home, school, farm, market, field, library, or community that uses this idea.

Glossary

Use this glossary to revise important words in Science and Technology.

- Living 1.3 Human Digestive System Things and their Environment: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.
- Animals: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.
- Plants 1.0 Living Things and their Environment: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.
- Matter 2.1 Properties: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.
- Force 3.3 Heat: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.
- Light: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.
- Force and its Effect 3.0 Force: A science idea to explain with observation, labelled diagrams, safe investigation, evidence, and a conclusion.

Add five more words from your lessons, then write a meaning and one correct example sentence or worked example for each word.

Answer And Teacher Notes

A strong Science and Technology answer should:

- answer the exact question asked;
- separate observation from explanation;
- name materials, variables, safety checks, or data sources where needed;
- use labelled diagrams, tables, or step-by-step procedures when useful;
- draw a conclusion from evidence, not from guessing.

For investigations, good answers include a question, method, results, conclusion, and one limit or improvement. For technology tasks, good answers explain the design choice and how it solves the problem safely.

Answer-key guide: a complete science answer names the concept, shows evidence in a labelled sketch/table/model, explains the pattern, includes safety where relevant, and ends with a conclusion beginning from the evidence. Wrong answers often mix observation with opinion or leave out variables.

Rubric for self-checking:

- 4: The answer addresses the exact question, uses subject vocabulary correctly, gives evidence or working, and includes a useful check or correction.
- 3: The main answer is correct but one part needs stronger evidence, clearer working, better labels, or a fuller explanation.
- 2: Some understanding is shown, but the answer is incomplete, too general, or hard to follow.
- 1: The answer is guessed, off topic, or missing the evidence needed for the task.

Correction routine: reread the command word, underline the key data or evidence, improve one weak step, and write one sentence explaining why the final answer is reasonable.

Final Project

Create a final Science and Technology project.

Your project must include:

1. A clear question, problem, or design need.
2. Evidence from at least three lessons.
3. A safe method, labelled diagram, data table, flow chart, model, or prototype plan.
4. A conclusion that explains what the evidence shows.
5. One safety note, one limitation, and one improvement.

Present your project to a classmate, teacher, parent, or study group.